



FRICTION -01

Class 08 - Science

Section A

1. Which of the following living organisms have streamlined body? [1]
 - a) Dog and horse
 - b) Fish and birds
 - c) Lion and Donkey
 - d) Elephant and human
2. Sound can travel through [1]
 - a) solids, liquids and gases
 - b) liquids only
 - c) solids only
 - d) gases only
3. It is difficult to walk on an oily floor because [1]
 - a) Force of friction is very less
 - b) Force of friction is absent
 - c) Force of friction is very high
 - d) Floor get spoiled
4. Sonic boom is due to [1]
 - a) Friction between air and supersonic object
 - b) Gravitational Force between Earth and plane
 - c) Friction between air and gravitation
 - d) Attraction between earth and atmosphere
5. Friction depends upon [1]
 - a) Smoothness of surface and nature of surface
 - b) Time of the activity
 - c) Smoothness of surface only
 - d) Nature of surface only
6. Oils and lubricants help in reducing friction because they [1]
 - a) Make the surface coarse
 - b) Increase the rolling friction
 - c) Fill the irregularities in surface
 - d) Decrease the rolling friction
7. Which of the following is an example of contact force? [1]
 - a) Frictional force
 - b) Electrostatic force
 - c) Magnetic force
 - d) Gravitational force
8. Fluids are [1]
 - a) Liquids and gases
 - b) Semisolids only
 - c) Gases only
 - d) Liquids only
9. A car with worn out treads on tyres is more likely to skid due to [1]

- a) Smaller size of tyres
b) Less friction
c) Less weight of tyres
d) More friction

10. In which of the substance friction is less [1]
a) Kerosene
b) Water
c) Air
d) Ice

11. Which of the following objects have low friction? [1]
a) Goalkeeper gloves
b) Wicket keeper gloves
c) Car tyres
d) Ice skates

12. Moving ball slowly decrease its speed in a grassy field due to [1]
a) Friction between ball and grass
b) Static force of ball
c) Rolling friction due to grass only
d) Gravitational force of earth

13. Which of the following method is not used to reduce friction? [1]
a) Greasing
b) Using ball bearing
c) Treading the tyres of vehicles
d) Lubricating

14. Force of friction always acts on all [1]
a) Heavy object
b) Lighter object
c) Stationary object
d) Moving object

15. In a large commercial complex, there are four ways to reach the main road. One of the paths has loose soil, the second is laid with polished marble, the third is laid with bricks and the fourth has a gravel surface. It is raining heavily and Paheli wishes to reach the main road. The path on which she is least likely to slip is [1]
a) bricks
b) loose soil
c) gravel
d) polished marble

16. Friction is increased by _____ [1]
a) powder sprinkled on carrom board
b) smoothening the surface
c) treading of tyres
d) lubrication

17. Whenever the surfaces in contact tend to move or move with respect to each other, the force of friction comes into play [1]
a) only if one of the two objects is liquid
b) irrespective of whether the objects are solid, liquid, or gaseous.
c) only if one of the two objects is gaseous.
d) only if the objects are solid.

18. The shoes of football player have spikes in their sole to [1]
a) Make frictionless
b) Increase friction
c) Decrease friction
d) Hard hitting the ball

19. The force of friction which comes into play when one body moves over the surface of other body is called [1]
a) Sliding friction
b) Rolling friction

- c) Kinetic friction
d) Static friction

20. Striker moves longer distance on carom when [1]

a) Striker is old
b) Striker is new
c) Powder is sprinkled
d) Carom board is neat and clean

21. It is difficult to move a cycle with brakes on because [1]

a) Brakes make the cycle frictionless
b) Friction increases
c) Friction decreases
d) Brakes increase the mass

22. When we attempt to move any surface, we apply a force to overcome [1]

a) Interlocking between surfaces
b) Gravitational force of earth
c) Electrostatic force between particles
d) Vander wall force between surfaces

23. What does air resistance do to falling objects? [1]

a) Increase its mass
b) Make them rise
c) Slows down them
d) Speeds them up

24. The surface of mortar and pestle (silbatta) used for grinding is etched again after long use to _____ friction. [1]

a) increase
b) decrease
c) remove
d) keep same

25. Which of the following method is applied in moving parts of machine to reduce friction? [1]

a) Converting sliding friction into rolling friction
b) Polishing the area of contact
c) Using oils and lubricants
d) Converting rolling friction into sliding friction

26. In machine, friction between moving part is reduced by [1]

a) Using lubricants
b) Using powder
c) Heating machine
d) Using paints

27. A boy rolls a rubber ball on a wooden surface. The ball travels a short distance before coming to rest. To make the same ball travel longer distance before coming to rest, he may [1]

a) sprinkle talcum powder on the wooden surface.
b) cover the ball with a piece of cloth.
c) sprinkle sand on the wooden surface.
d) spread a carpet on the wooden surface.

28. Rolling friction and sliding friction are two types of [1]

a) Static friction
b) Directional friction
c) Dynamic or kinetic friction
d) Fluid friction

29. The force responsible for changing the state of motion of an object is [1]

a) Molecular force
b) Muscular force
c) Friction
d) Gravitational force

30. Rolling friction is smaller than? **[1]**

 - Fluid friction
 - All of these
 - Static friction
 - Sliding friction

31. Which shape will have minimum friction in a fluid? **[1]**

 - Parabolic tips
 - Rectangular tips
 - Pointed tips
 - Circular tips

32. The frictional force exerted by fluids is also called **[1]**

 - Drag
 - Smug
 - Slag
 - Rung

33. What kind of substances are known as lubricants **[1]**

 - None of these
 - Decrease friction
 - Increase or decrease friction
 - Increase friction

34. A force which opposes the motion is called **[1]**

 - Linear
 - Friction
 - Sliding
 - Gravitational

35. To walk on slippery ground is difficult because the **[1]**

 - Ground offer very high resistance to walk
 - Habit of walking on slippery ground is not enough to us
 - Frictional force is not enough to prevent slip
 - Frictional force is very high to walk

36. Friction produces **[1]**

 - Force
 - Motion
 - Heat
 - Light

37. Friction offered by water is good for the **[1]**

 - Movement of ships or boat
 - Respiration of aquatic organisms
 - Health
 - Breathing of aquatic plants

38. Streamlining is **[1]**

 - making back thick
 - making back thin
 - Making tips thin
 - making tips thick

39. The frictional force that counter balance the applied force during stationary state is called **[1]**

 - Static friction
 - Kinetic friction
 - Momentary friction
 - Rolling friction

40. Carpenters often grease screws before fixing them to **[1]**

 - Increase friction
 - Maintain friction
 - Decrease friction
 - Make it light weight

41. A toy car released with the same initial speed will travel farthest on **[1]**

- a) cemented surface
b) brick surface
c) muddy surface
d) polished marble surface

42. The handle of a cricket bat is usually rough to _____. **[1]**
a) get ball on bat
b) decrease friction
c) increase friction
d) get more run

43. Ball bearing is used in cycles because **[1]**
a) Rolling friction is more than sliding friction
b) Rolling friction is less than sliding friction
c) Ball bearing increase the wheel size
d) Ball is used to make attractive

44. Mobil oil is applied on iron shutters to _____ friction. **[1]**
a) keep same
b) increase
c) decrease
d) remove

45. Friction involves **[1]**
a) Unnecessary expenses of energy
b) Increase the life of machine
c) Reduce the production
d) Reduce the expenses of energy

46. Friction between any two objects is due to **[1]**
a) Even surfaces of them
b) Attraction between them
c) Repulsion between them
d) Surface irregularities on them

47. In moving machine, friction **[1]**
a) Decrease the energy loss
b) Increase the cost of machine
c) Reduce the efficiency of machine
d) Increase the efficiency of machine

48. When the friction is more **[1]**
a) One body is rough and other is smooth
b) Both bodies are in liquid state
c) Two bodies in contact are smooth
d) Two bodies in contact are rough

49. The heat in metals is conducted by the process of **[1]**
a) Conduction
b) Absorption
c) Radiation
d) Convection

50. Frictional force is necessary in **[1]**
a) Playing carom
b) Accelerating the car
c) Using matchstick
d) Pushing heavy load

51. Sprinkling of powder on the carom board **[1]**
a) Maintain the friction
b) Reduce friction
c) Increase friction
d) Make the board durable

52. A body is rolling over the ground or sliding over the same ground. In which case, the friction will be more? **[1]**
a) When sliding
b) When rolling

- c) Equal in both case
d) There is no friction in both case

53. Fluid friction can be minimized by **[1]**
a) Adding salts to fluid
b) Giving suitable shape
c) Increasing volume
d) Decreasing volume

54. Which of the following method is applied to increase friction of vehicles? **[1]**
a) Regular oiling of engine
b) Less inflating the tyres
c) Cleaning of vehicles
d) Treading of tyres

55. It is difficult to apply brakes to stop the moving ship because **[1]**
a) There is no friction between solid and liquid
b) Friction exerted by water is very small
c) Friction exerted by water is very high
d) Friction exerted by water is absent

56. To sharpen the blade of a knife by rubbing it against a surface, which of the following will be most suitable? **[1]**
a) stone
b) wooden block
c) glass block
d) plastic block

57. The sole of shoes becomes flat after using few months because of **[1]**
a) Wearing due to bad quality
b) To buy new shoes
c) Wearing due to friction
d) Wearing due to rough weather

58. Sandpaper is used to clean the metal surface to remove rust as because **[1]**
a) Sandpaper has rough surface to increase friction
b) Sand paper has smooth surface to increase friction
c) Sand paper has rough surface to decrease friction
d) Metal and sand paper reacts easily

59. Friction of air can be observed if a falling object is **[1]**
a) Light
b) Small
c) Heavy
d) Large

60. Friction is a **[1]**
a) Contact force acting in same direction
b) Non-contact force in same direction
c) Non-contact force in opposite direction
d) Contact force acting in opposite direction

61. The SI unit of frictional force is **[1]**
a) Newton
b) Watt
c) Joule
d) Pascal

62. If we apply oil on door hinges, the friction will **[1]**
a) increase
b) decrease
c) will remain unchanged
d) disappear altogether

63. The force required to keep the object moving with same speed is a measure of **[1]**
a) Rolling friction
b) Sliding friction

c) Static friction

d) Contact friction

Section B

64. In Fig. a boy is shown pushing the box from right to left. The force of friction will act on the box [1]



a) from right to left ($\leftarrow$)

b) from left to right ($\rightarrow$)

c) vertically upwards ($\uparrow$)

d) vertically downwards ($\downarrow$)

Section C

65. The direction of friction is always _____ to the direction in which the object is moving. [1]

a) Equal

b) Perpendicular

c) Unrelated

d) Opposite

66. Sliding friction is _____ than static friction. [1]

a) More

b) Less

c) Greater

d) Equal

67. Friction opposes the _____ between two surfaces in contact. [1]

a) Reflective motion

b) Relative motion

c) Refractive motion

d) Rotary motion

68. Friction depends on the _____ of surface. [1]

a) Matter

b) Smoothness/ nature

c) Size

d) Colour

69. A force of _____ acts when an object moves through air or water. [1]

a) Reacceleration

b) Acceleration

c) Weight

d) Friction

Section D

70. Which of the following is correct order in descending frictional force [1]

a) Static > sliding > rolling

b) Rolling > static > sliding

c) Static > rolling > sliding

d) Rolling > sliding > static

Section E

71. Which of the following statements is incorrect? [1]

a) Friction acts on a ball rolling along the ground.

b) Friction acts on a boat moving on water.

c) Friction does not act on a ball moving through air.

d) Friction acts on a bicycle moving on a smooth road.

72. Four children were asked to arrange forces due to rolling, static and sliding frictions in a **decreasing order**. [1]
Their arrangements are given below.
Choose the correct arrangement.
- | | |
|-----------------------------|-----------------------------|
| a) static, sliding, rolling | b) sliding, static, rolling |
| c) rolling, sliding, static | d) rolling, static, sliding |

Section F

73. Alida runs her toy car on dry marble floor, wet marble floor, newspaper and towel spread on the floor. The force [1]
of friction acting on the car on different surfaces in **increasing order** will be
- | | |
|---|--|
| a) wet marble floor, dry marble floor, towel,
newspaper. | b) wet marble floor, dry marble floor,
newspaper and towel. |
| c) newspaper, towel, dry marble floor, wet
marble floor. | d) towel, newspaper, dry marble floor, wet
marble floor. |